

CSE422: Artificial Intelligence Lab

Lab Project Report

Can Routine Health-Screening Data Predict Diabetes?

An Investigation of 200,000 Student Records: Imbalance, Overlapping Data Leakage, and Model Evaluation

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September 9, 2026

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An Investigation of 200,000 Student Records: Imbalance, Overlapping Data Leakage, and Model Evaluation

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Abstract—Type-2 diabetes is frequently undiagnosed in young adults due to asymptomatic early stages and lack of routine screening. This project shows whether standard health measurements collected during admission can predict diabetes status efficiently. Analyzing a 200,000-row dataset, we uncovered a critical flaw: 100,000 records were repeated and masked, causing extreme data leakage across splits. After merging the overlapping records and recovering 90% of missing values, five supervised models (Neural Network, Logistic Regression, KNN, Decision Tree, Naive Bayes) and an unsupervised K-Means model were evaluated. Despite reaching 90% accuracy due to class imbalance, all models scored an ROC-AUC of 0.50 due to the accuracy paradox.

Index Terms—Classification, data leakage, class imbalance, missing value imputation, neural networks, k-means clustering, accuracy paradox.

I. INTRODUCTION

TYPE-2 diabetes is often undetected in young adults because it is typically asymptomatic during early stages, and routine screening is uncommon at university age. If routine measurements taken upon university admission (such as age, sex, height, weight, blood type, temperature, heart rate, blood pressure, cholesterol, and smoking habits) could predict diabetes, health centers could direct limited diagnostic testing budgets toward the highest-risk individuals.

A. Target Feature

We selected a feature with discrete values as our target feature: **Diabetes**, a binary categorical feature taking the values `Yes` and `No`.

B. Aim and Motivation

The aim of this project is to build and evaluate machine learning models to solve the screening classification problem. The motivation is to determine if basic non-invasive measurements carry sufficient statistical signals to predict a student's diabetes risk without requiring expensive upfront lab tests.

II. DATASET DESCRIPTION

A. Classification Problem

This is a **classification problem** because the target feature (**Diabetes**) takes discrete categorical values (`Yes` or `No`) rather than continuous numerical values.

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Submitted for CSE422: Artificial Intelligence.

B. Dataset Size and Features

The raw dataset contains 200,000 rows and 13 columns. The `Student ID` feature is a non-predictive identifier and was dropped, leaving 11 predictive features. Table I details the raw dataset's structure.

TABLE I
STRUCTURE OF THE RAW DATASET

Property	Value
Total Records (Rows)	200,000
Total Attributes (Columns)	13
Predictive Features	11
Identifier Column	Student ID (Dropped)
Target Feature	Diabetes (Binary: Yes/No)
Missing Values	Exactly 10.0% in every column
Exact Duplicate Rows	7,644
Overlapping / Repeated Records	100,000 (Two masked copies)

C. Types of Features

The dataset consists of 8 quantitative and 3 categorical features:

- **Quantitative (8):** Age, Height, Weight, BMI, Temperature, Heart Rate, Blood Pressure, Cholesterol.
- **Categorical (3):** Gender (Binary), Smoking (Binary), Blood Type (Nominal: A, B, AB, O).

D. Dataset Balance / Imbalance

For the output target feature, all unique classes do **not** have an equal number of instances. The dataset is heavily imbalanced with a 9:1 ratio:

- `No` (Non-diabetic): 90.0% of labeled instances.
- `Yes` (Diabetic): 10.0% of labeled instances.

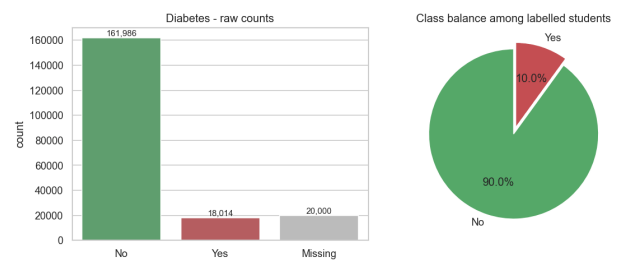


Fig. 1. Distribution of the target feature (**Diabetes**). The 9:1 class imbalance demonstrates a majority-class baseline accuracy of 90%.

Fig. 1 illustrates this class distribution across the $N = 2$ classes, as required by the project specifications.

III. DATA ANALYSIS

A. Exploratory Data Analysis (EDA)

We performed exploratory data analysis to evaluate feature distributions and detect anomalies. Marginal distributions revealed that six quantitative features (Age, Height, Weight, Heart Rate, Blood Pressure, and Cholesterol) have completely flat, uniform distributions between strict numerical boundaries (Fig. 2).

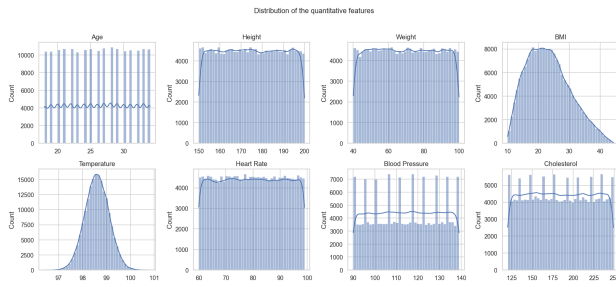


Fig. 2. Histograms of quantitative features displaying uniform distributions bounded strictly between artificial limits.

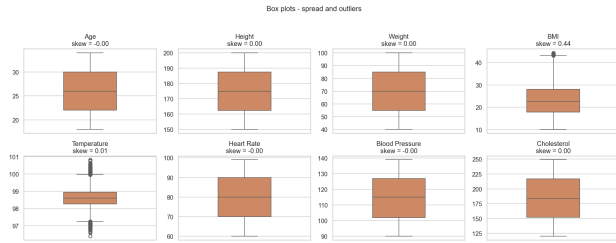


Fig. 3. Box plots of quantitative features showing zero outliers beyond the whiskers, indicative of synthetic data generation.

Furthermore, box plot analysis (Fig. 3) showed that not a single outlier exists outside the whiskers across any quantitative attribute. Real physiological data inherently exhibits long-tail outliers; their absence strongly suggests synthetic generation.

B. Correlation Test Analysis

Fig. 4 presents the Pearson correlation matrix for all quantitative features.

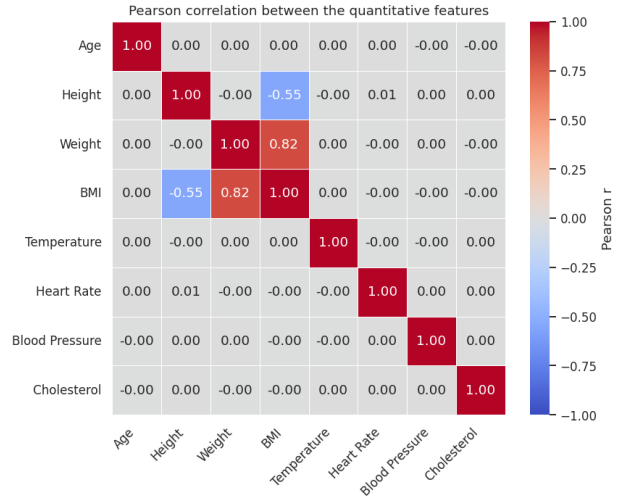


Fig. 4. Pearson correlation heatmap across quantitative features.

Two major conclusions were reached from the correlation test:

- 1) All independent features show zero correlation with one another ($|r| < 0.01$). Even Height and Weight share zero correlation, which is biologically unnatural.
- 2) BMI shows severe correlation with Height and Weight because it is an exact mathematical duplicate derived from $\text{Weight}/(\text{Height}/100)^2$.

IV. DATASET PRE-PROCESSING

To satisfy the guidelines, we discuss each observed problem along with the corresponding solution and applied pre-processing technique.

A. Problem 1: 100,000 Overlapping Rows and Data Leakage

Problem: The dataset contained 200,000 rows, but Student ID only ranged from 1 to 100,000. The file was actually two vertically stacked copies of 100,000 student records, where different random 10% subsets of cells were blanked out in each copy. Because the masked nulls differed, standard deduplication (`drop_duplicates()`) caught only 7,644 rows. Splitting this raw file causes extreme data leakage across train and test sets, falsely inflating gradient boosting AUC from 0.50 to 0.58.

Solution: We merged the two halves using `combine_first`. This eliminated duplicate leakage completely, collapsed the dataset back to the true 100,000 unique records, and recovered roughly 90% of all missing cells simultaneously.

B. Problem 2: Unlabeled Target Rows

Problem: After merging, 1,007 rows still had a null Diabetes label.

Solution: These rows were dropped entirely. Imputing target labels introduces false assumptions and invalidates model evaluation.

C. Problem 3: Handling Remaining Missing Values

Problem: Approximately 1% of feature cells remained blank after merging.

Solution: Missing values were imputed based on feature type:

- **Quantitative features:** Imputed using the **Median**, which is robust against skewness.
- **Categorical features:** Imputed using the **Mode** (most frequent category).

D. Problem 4: Encoding Categorical Features

Problem: Models require numeric arrays, but Gender, Smoking, and Blood Type were text strings.

Solution:

- Target Diabetes was label-encoded (Yes = 1, No = 0).
- Binary predictors (Gender, Smoking) were one-hot encoded with `drop_first=True` to prevent dummy variable traps.
- Nominal predictor Blood Type (A, B, AB, O) was one-hot encoded with all four indicator columns to avoid imposing an artificial order on distance-based models like KNN.

E. Problem 5: Feature Selection and Multicollinearity

Problem: BMI is perfectly collinear with Height and Weight, inflating the variance of logistic regression coefficients.

Solution: Mutual information and correlation thresholding confirmed that BMI provided zero unique information. Dropping it removed multicollinearity without reducing predictive performance.

F. Problem 6: Feature Scaling

Problem: Features like Cholesterol (120–250) have much larger magnitudes than Temperature (97–101), which distorts Euclidean distance calculations in KNN and slows neural network convergence.

Solution: We applied Standard Scaling (Z-score normalization, Eq. 1) to all features:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

The scaler was fitted strictly on the training partition only to prevent data leakage.

V. DATASET SPLITTING

To avoid class imbalance bias across splits, we used a **Stratified Split** (`stratify=y`) with a fixed random seed (`random_state=42`).

- **Train Set (80%):** 79,194 records used for training. (A 20% validation subset was temporarily created from this for Neural Network early stopping).
- **Test Set (20%):** 19,799 held-out records used exclusively for testing and final metric reporting.

Both sets preserved the identical 10.0% positive diabetes proportion.

VI. MODEL TRAINING & TESTING (SUPERVISED)

We implemented and evaluated five supervised learning models:

- **K-Nearest Neighbors (KNN):** Non-parametric spatial classifier configured with $k = 15$ on standardized inputs.
- **Decision Tree:** Tree classifier with `max_depth=5`. Tested in both unweighted and class-weighted modes.
- **Logistic Regression:** Linear classifier modeling log-odds. Tested in unweighted and class-weighted modes.
- **Gaussian Naive Bayes:** Probabilistic classifier assuming conditional independence among features.
- **Neural Network:** Multi-Layer Perceptron built using Keras with an input layer (13 encoded features), two hidden layers (32 and 16 ReLU units) with 0.2 Dropout, and a single Sigmoid output unit. Trained with binary cross-entropy, Adam optimizer ($lr = 0.001$), batch size 256, and early stopping.

VII. MODEL SELECTION AND COMPARISON ANALYSIS

Table II and Fig. 5 present the complete comparison of all supervised models across Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

TABLE II
MODEL PERFORMANCE COMPARISON ON THE TEST SET

Model	Accuracy	Precision	Recall	F1-Score	AUC
Neural Network	0.9000	0.0000	0.0000	0.0000	0.4760
NN (Class-Weighted)	0.6664	0.0964	0.2788	0.1432	0.4850
Logistic Regression	0.9000	0.0000	0.0000	0.0000	0.4869
LogReg (Balanced)	0.5019	0.0966	0.4768	0.1607	0.4861
KNN ($k = 15$)	0.9000	0.0000	0.0000	0.0000	0.5012
Decision Tree ($d = 5$)	0.8996	0.0000	0.0000	0.0000	0.4945
Decision Tree (Balanced)	0.3883	0.0970	0.6157	0.1676	0.4869
Gaussian Naive Bayes	0.9000	0.0000	0.0000	0.0000	0.4857
Majority Baseline	0.9000	0.0000	0.0000	0.0000	0.5000

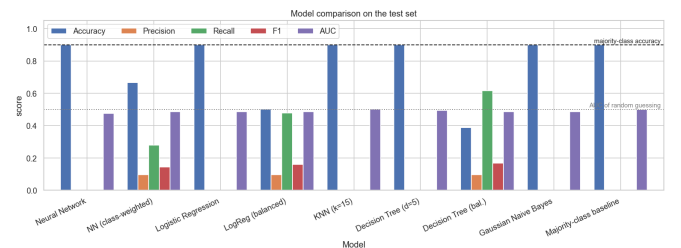


Fig. 5. Bar chart comparing prediction accuracy and evaluation metrics across all models.

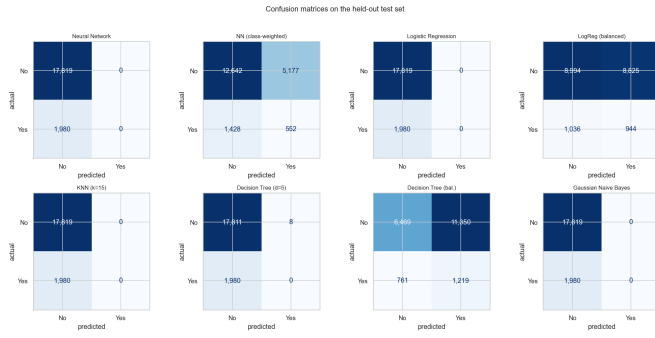


Fig. 6. Confusion matrices for all models on the held-out test set.

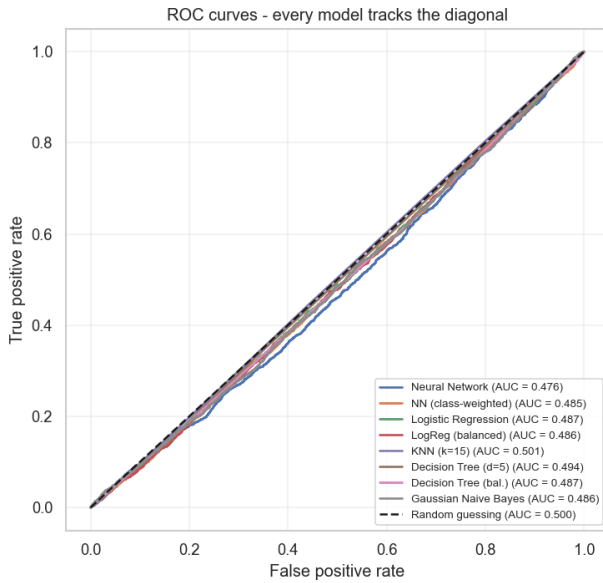


Fig. 7. ROC curves and AUC scores for each evaluated model.

A. Metrics Analysis and Confusion Matrix

All unweighted models achieved 90.0% accuracy simply by predicting the majority class (No) for every test instance. Consequently, their Precision, Recall, and F1-Scores are exactly 0.0000 (Table II). The confusion matrices (Fig. 6) confirm that the unweighted models have a completely empty predicted positive column.

Class-weighting raised Recall (up to 0.6157 for Decision Tree), but dropped Accuracy to 38.8%–66.6% while Precision remained pinned at 10% (the base prior rate).

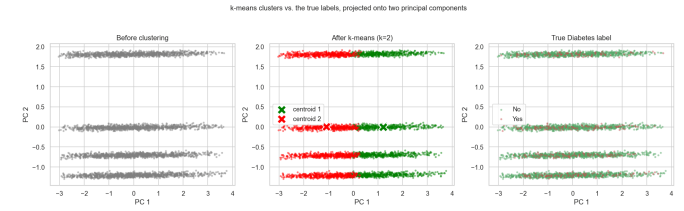
B. ROC Curves and AUC

Fig. 7 shows that all ROC curves lie directly on the diagonal random-guess line, with AUC scores hovering around 0.50. This proves that despite high accuracy, no model possessed true predictive power.

VIII. UNSUPERVISED LEARNING

We removed the target label (Diabetes) from the dataset to treat it as an unsupervised task. We then applied **K-Means**

Clustering ($k = 2$) on the standardized feature space to see if natural geometric clusters aligned with diabetic and non-diabetic groups.

Fig. 8. K-Means clustering ($k = 2$) on the unlabeled data compared against true labels.

As shown in Fig. 8, the feature space forms an undifferentiated continuous cloud. K-Means cut the cloud arbitrarily along the axis of highest variance. Comparing the $k = 2$ clusters against the true labels yielded an Adjusted Rand Index (ARI) of -0.00003 , confirming that unsupervised clustering cannot detect diabetes categories from these features.

IX. CONCLUSION

A. Best Model Selection

Judged by ROC-AUC, no model outperformed random guessing (all ≈ 0.50). However, **Logistic Regression** is the most suitable model to report: it matches the highest baseline accuracy (90.0%), trains in under one second, and its linear coefficients make the absence of predictive signal immediately transparent rather than obscuring it inside a black box.

B. Potential Reasons for Low Performance

The primary reason for poor performance is that the dataset is synthetic, with features generated independently of the diabetes label. Mutual information across all features was below 0.012 nats, effect sizes (Cohen's d) were under 0.022, and diabetes rates were completely flat at 10% across every decile of every feature. No algorithm can extract signal where none exists.

C. Challenges Faced

- 1) **100,000 Repeated/Overlapping Records:** Spotting that the file consisted of two masked copies of 100,000 records was difficult because `drop_duplicates()` failed to catch non-identical masked cells. Merging them was essential to avoid severe leakage.
- 2) **9:1 Class Imbalance and Accuracy Paradox:** The severe imbalance initially gave a deceptive 90% accuracy while models predicted zero positive cases. Resolving this required implementing ROC-AUC and class weighting.

REFERENCES

- [1] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.
- [2] R. J. A. Little and D. B. Rubin, *Statistical Analysis with Missing Data*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2019.

- [3] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 1–21, 2012.
- [4] F. Chollet *et al.*, “Keras,” 2015. [Online]. Available: <https://keras.io>.
- [5] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. 3rd Int. Conf. on Learning Representations (ICLR)*, San Diego, CA, 2015.
- [6] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] L. Hubert and P. Arabie, “Comparing partitions,” *Journal of Classification*, vol. 2, no. 1, pp. 193–218, 1985.